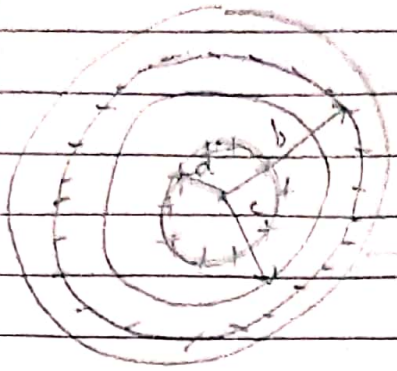
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No.
Date:



Kapasitor Bola

$$E = k_e \frac{q}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2}$$

$$V = -\int E\,dr = -\frac{q}{4\pi\varepsilon_0}\int_b^a \frac{dr}{r^2} = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\,\frac{b-a}{ab}$$

Subsitusi ke persamaan

$q = CV$

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a}$$ (kapasitor berbentuk bola)

Kapasitansi

$$C = 4\pi\varepsilon_0 \frac{a \cdot 1}{1 - a/b}$$

$b \to \infty$ dan R menjadi a

$C = 4\pi\varepsilon_0 R$ (bola terisolasi)

$$E = \left(\frac{q}{4\pi\varepsilon_0 R^3}\right) r \qquad V = -\int_b^a \vec{E} \cdot \vec{ds}$$

$$V = -\int_b^a \left(\frac{q}{4\pi\varepsilon_0 R^3}\right) r \cdot \vec{ds}$$

Dipindai dengan CamScanner